

Report 234

Record Sum Diagonal Expansions and Integer Inverses

Complete cancellation of the nonoscillatory correction sector

5 October 2026

Abstract

Let a_n count permutations of $\{1, \dots, n\}$ whose cycle minima sum to n . The same numbers count permutations whose record positions sum to n and form OEIS A368246. For $b_n = a_n/(n-1)!$, we prove a finite expansion to every fixed inverse-power order. The constant is $e^{-\gamma}$, the term of order n^{-1} is absent, and every subsequent coefficient is a polynomial in $\log n$ with periodic coefficients of mean zero. The reason is the exact rooted-tree identity $T(se^{-s}) = s$, which cancels all higher logarithms at the positive singularity of a shifted infinite product. We give the second- and third-order terms, Gamma-product root amplitudes, the exceptional delay caused by zeros at $\pm i$, a direct global boundary remainder, and a real finite Newton construction with a rigorous two-ceiling inverse envelope. The method and the leading constant are classical; the focus is the sequence-specific cancellation and its refinements. Numerical checks are separated from the analytic proof, and no effective remainder constant, convergence of an infinite transseries, or publication-priority claim is made.

[write] **Status and trust boundary, added 7 October 2026, batch 113.** This is bundle Report 234 of an external AI-assisted research session, filed in ProveIt as the single-source research report `a368246-cycle-minima-sums` (provenance, the sources as the write read them, what was checked, relation to the repository, the collected non-claims and the reading conventions are in Section 1.4). It is unrefereed and not formalized. Its title page and PDF author field read “Report 234”: it names no person or tool. *What rests on what.* Every theorem has a conventional proof in the text: an exact counting identity, the rooted-tree identity $T(se^{-s}) = s$, integer-polylogarithm and Gamma-product expansions at the roots of unity, a differentiated boundary remainder with Fourier integration by parts, exact coefficient transfer, Stirling’s formula and Newton’s method. The leading constant is also a consequence of the local limit theorem of Giuliano, Szewczak and Weber, which the write read (Remark 11.1). No proof uses a computation; the exact checks corroborate finite identities and the floating evaluations are illustrations. The write adds Remarks 9.3 and 11.1 and dated notes; no statement, proof or number of the source was changed.

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1 The expansion and its scope

For $n \geq 1$, let a_n be the number of permutations of $[n] = \{1, \dots, n\}$ for which the sum of the least elements of all cycles is n . The empty permutation is counted separately by $a_0 = 1$. The initial values are

$$1, 1, 0, 2, 3, 8, 90, 384, 2940, 18864, 232848, 1919520, \dots,$$

starting at $n = 0$. Put

$$b_n = \frac{a_n}{(n-1)!} \quad (n \geq 1), \quad A = e^{-\gamma}, \quad (1.1)$$

where γ is Euler's constant. The normalization is never applied at $n = 0$.

Theorem 1.1 (Fixed order expansion). *For every fixed integer $K \geq 2$ there are real functions $P_j(n, u)$, $2 \leq j \leq K$, such that*

$$b_n = A + \sum_{j=2}^K \frac{P_j(n, \log n)}{n^j} + \mathcal{O}_K\left(\frac{(1 + \log n)^{K-1}}{n^{K+1}}\right) \quad (n \rightarrow \infty). \quad (1.2)$$

Each $P_j(n, u)$ is a polynomial in u of degree at most $j - 2$. Its coefficients are periodic in the integer n , with period dividing

$$\lambda_j = \text{lcm}(2, 3, \dots, j),$$

and each coefficient has mean zero over that period. In particular, no nonoscillatory algebraic correction occurs at any fixed order, and there is no term of order n^{-1} .

The quantifier “fixed” matters. The implied constants and the sufficiently large starting index may depend on K . The theorem does not assert convergence when $K \rightarrow \infty$, a uniform estimate in K , or a summation of an infinite transseries. It describes a compatible hierarchy of finite asymptotic expansions. Polynomial growth in $\log n$ is included explicitly in the error.

Let $\omega = \exp(2\pi i/3)$ and define

$$C_\omega = \prod_{r=1}^3 \frac{\Gamma(r/3)}{\Gamma((r + \omega^{r+1})/3)}. \quad (1.3)$$

Theorem 1.2 (The first two corrections). *The expansion through order three is*

$$b_n = A + \frac{(-1)^n}{n^2} + \frac{(-1)^n(8 - 2\gamma - 2\log n) + 6 \operatorname{Re}(C_\omega \omega^{1-n})}{n^3} + \mathcal{O}\left(\frac{(1 + \log n)^2}{n^4}\right). \quad (1.4)$$

For orientation only, a high precision evaluation gives

$$C_\omega \approx 0.83150170079289916148 - 1.03642907874318733151 i.$$

The displayed decimal is an arithmetic approximation, not an interval certificate. The exact definition is (1.3).

[write] **The printed decimal, 7 October 2026.** The write evaluated (1.3) at 50 digits: $C_\omega = 0.8315017007928991614835830\dots - 1.0364290787431873315053703\dots i$, equal to an independent evaluation of the partial products to twelve digits. The printed real part is a truncation; the printed imaginary part is the rounding of $-1.03642907874318733150\dots$, whose truncation at the printed length ends in $\dots 33150$.

1.1 Why the positive singularity is not enough

The exact generating function is

$$F(z) = \prod_{j \geq 1} \left(1 + \frac{z^{j+1}}{j}\right), \quad b_n = [z^{n-1}]F(z), \quad |z| < 1. \quad (1.5)$$

Its unit circle is a natural boundary. The pole at 1 supplies the constant A , but the nontrivial roots of unity supply every algebraic correction. A radial estimate as $z \uparrow 1$ cannot, by itself, prove (1.2). In particular, subtracting the positive pole does not produce analytic continuation across a whole arc around 1.

We instead split F into a finite polylogarithmic factor and a tail with as many bounded boundary derivatives as the chosen order requires. Only finitely many roots then need explicit local subtraction. A global absolutely continuous boundary trace justifies repeated Fourier integration by parts. This establishes an actual coefficient estimate, including the contribution of all the unselected high-order roots.

1.2 Attribution and the bounded novelty claim

The finite record polynomial is classical and appears explicitly in Louchard [4, Eq. (1)]; its conversion to (1.5) is elementary. The leading constant A follows immediately from the Dickman local limit

theorem of Giuliano, Szewczak and Weber [5, Theorem 2.1]. The general hybrid method, including expansions at roots of unity despite a natural boundary, belongs to Flajolet, Fusy, Gourdon, Panario and Pouyanne [7]. Their Section 3 analyzes the related product $\prod_{j \geq 1} (1 + z^j/j)$, whose correction sector is different from that of (1.5).

The specific calculation developed here is the cancellation of the entire nonoscillatory algebraic sector for the shifted exponent $j + 1$, together with explicit oscillatory coefficients and a justified integer inverse. These particular refinements were not found in the sources inspected for this report. That statement is a bounded literature observation, not a claim of worldwide priority. Section 11 describes the source and duplicate-search limits, including the lack of a freshly retrieved OEIS revision for A368246.

1.3 Guide to the proof

Section 2 fixes the combinatorial normalization and index shift. Sections 3–5 construct the local expansions with differentiated remainders. Section 6 supplies the global boundary argument, and Section 7 converts it into Theorem 1.1. Section 8 proves Theorem 1.2 without fitting coefficients. Section 9 gives the threshold envelope and a finite Newton approximation to its explicit model root. The final sections document the reproducible arithmetic, known limitations, and sources, then propose further research questions.

1.4 Provenance, sources and reading conventions

[write] **Added 7 October 2026, batch 113.** *Provenance.* This report is Report 234 of the session bundle that ProveIt’s docs/incoming intake processed as batch 113: the archive Report234.zip (648,517 bytes, 25 files [independent check, 7 October 2026: 24 files; the archive listing has 24 entries, all under the wrapper], wrapper directory Report234/), arrival commit 60f54ea06, placed in a4186a946. The text is the delivered article.tex (67 lines) with its eleven \input files sections/01_results.tex, ..., sections/11_sources.tex (29 pages, dated 5 October 2026), shipped under the same names. The programs are shipped under code/ (the delivered root build.py as code/build.py); the receipts code/receipt.json and code/guard_receipt.json and the requirements under data/; SOURCES.md as delivered. Not shipped, and retrievable from the arrival commit: the delivered Report234.pdf, the checksum manifest MANIFEST.sha256 (23 entries, all verified at the write) and the delivered README.md, which the report README replaces. The package records no ProveIt commit, although SOURCES.md names five reports of this collection and the blob hashes of four of them (all four blobs are in the repository’s history: two are the current versions, two are earlier versions since revised). It carries no “prepared for private review” line and no personal data. It is a single-source report, so no merge choice was made. The write prefixed the 104 delivered labels with cms: and updated the references to them (75 \eqref, 27 \ref); it added this subsection, the notes marked [write] and Remarks 9.3 and 11.1, each the last statement of its section, and inserted no numbered display, so every section, theorem and equation of the source keeps its delivered number.

The sources, as the write read them (7 October 2026): OEIS A368246 and A143946 in the internal format, with the b-file of A368246 (Remark 11.1); Giuliano–Szewczak–Weber, arXiv:1309.1578v1 (37 pages), Theorem 2.1 and its model on p. 2, which is the model of Section 2; the transseries volume Analysis/Transseries/docs/series-and-transseries/Transseries_And_Inversion/transseries_and_inversion.tex (p0:def:model, p0:thm:lambert-core, p0:def:three-inverses, p0:thm:staircase, plt:thm:lw-template). Not read by the write: Louchard, Kortchemski, de la Bretèche–Tenenbaum and Flajolet–Fusy–Gourdon–Panario–Pouyanne; the source’s account of them (Section 11 and SOURCES.md) is reported, not confirmed, in particular the literal invocation of the hybrid theorem in Section 6.5.

What was checked at intake. The batch-113 dossier read the manuscript, found no demonstrably wrong claim,

recorded the local limit theorem as imported, and reran the delivered suite on a copy (it reproduces apart from a recorded Python version string). *What the write checked.* It read every proof and found no error. With its own code (Python 3.14.4, exact integers and fractions, 80-digit fixed point; SymPy 1.14.0; mpmath 1.3.0 at 30–60 digits), not importing the delivered programs, it

- computed a_n from (2.1) for $n \leq 451$, equal to all 452 terms of the OEIS b-file and to the displayed values; the shifted product (2.3) exactly for $n \leq 61$; and, for $n \leq 8$, both statistics on every permutation: the counts are a_n and the two distributions coincide;
- computed b_n for $n \leq 4001$ from (1.5) in 80-digit fixed point (agreeing with the exact values for $n \leq 451$ to 10^{-61}): the remainder of (1.4), times $n^4/(1 + \log n)^2$, has absolute value at most 5.52 for $200 \leq n \leq 4001$; $n^2(b_n - A) = \pm 1$ up to $O(\log n/n)$, and its means over six consecutive n are of size 10^{-6} to 10^{-8} , as the absence of an n^{-1} term requires;
- evaluated C_ω and Q_i by the Gamma products (5.5), (5.8) and independently by extrapolated partial products (twelve and ten digits agree), $F(-1) = 1$, the slope (8.3) ($S = 0.7413602942\dots = 2\log 2 + 1 - \zeta(2)$), the $r = 1$ slope $\frac{3}{2} - \log 2$ symbolically, and the sum $\frac{1}{2}$ of the four contributions;
- verified the tree cancellation (4.5) through degree 40 and the leading terms (5.3) for $2 \leq q \leq 14$ with its own exact coefficients (with $B_2 = s - 2s^2 + \dots$, $B_4 = \frac{8}{3}s^3 + \dots$);
- derived the jet (8.5) from (8.4) symbolically; verified (8.6) exactly for $3 \leq m \leq 30$, (8.7) against the exact coefficients at $m = 1000$ and 4000, the shift to (8.8), the generic coefficient (5.7) at $q = 3$ and the constant 256 of (5.10);
- checked (9.14), (9.15) ($24X$ times the residual tends to -1) and (9.16) for the model B_3 ($(r_3 - X - \frac{1}{2})X \log X \rightarrow \frac{1}{24}$), and the monotone quadratic convergence of (9.18), at $\log y = 10^3, 10^5, 10^8$;
- checked the claim on constant shifts in Section 12 numerically: $m(b_m^{(d)} - A)/(A(1 - d))$ is 0.9970 ($d = 0$) and 1.0017 ($d = 2$) at $m = 3000$.

It reran, on a fresh copy of the delivered layout, `checks.py` (normal and `-0`): both receipts equal `data/receipt.json` apart from the recorded Python version (3.14.4 against 3.12.14); `record_sum.py` with `exact` and `inverse` runs. The guard suite `guard_tests.py` stops on Windows at “file path guard 5”, a path written with forward slashes and a `..` component, which is not a mathematical failure. The PDF and ZIP builder and `reproduce_zip.py` were not run. Floating computations are observations, not certificates.

Relation to the repository. Before batch 113 no file of the repository named A368246 or A143946. The five reports that SOURCES.md inspected as possible duplicates (`a039831-two-fourier-peaks`, `a372395-acyclic-orientation-partitions`, `a279619-level-seven-gamma-constant`, `a238016-restricted-partitions-cubic-boundary`, `a189281-path-forest-expansions`) treat other models; no result is shared and no reciprocal note is proposed. No Lean or Rocq file treats these sequences; no statement of this report is formalized, and its place in the collection confers no formal status. The inverse is compared with the transseries volume in Remark 9.3.

[write] What is not claimed, collected from the source. The finite counting polynomial (Louchard, Eq. (1)), its reduction to the shifted product, the leading constant $e^{-\gamma}$ (Giuliano–Szewczak–Weber, Theorem 2.1), Gamma grouping at roots of unity and the hybrid method (Flajolet–Fusy–Gourdon–Panario–Pouyanne) are prior; the refinements were not found in the inspected sources, a bounded observation and no priority claim. Not claimed: convergence of an infinite root-of-unity transseries or uniformity in K ; effective constants or onsets; a finite threshold certificate from rounding a model root; interval certification of the printed Gamma amplitudes; a complete implementation of every higher P_j ; formal verification, peer review or a worldwide priority search; that the interpolation of Section 9 interpolates the sequence; that the guard tests prove security. *The write adds:* its checks are exact or floating computations; its transseries remark claims no novelty for any inversion.

Reading conventions (letters the source uses in more than one sense, with the tempting false readings; no symbol was renamed).

Symbol	Meanings
A, A_K	$A = e^{-\gamma}$; $A_K(x) = \Gamma(x)B_K(x)$ the real model (9.2).
B	$B_{1,L}, B_{q,L}, B_q(s)$ the (filtered) tree series of Sections 4–5; $B_K(x)$ the real interpolation of Section 9. <i>False reading</i> : $B_2(s) = s - 2s^2 + \dots$ in (8.2) and $B_2(x) = A + \cos(\pi x)/x^2$ in (9.3) are unrelated.
C, D	C_ω the Gamma amplitude; $C_K, \tilde{C}_K, D_K$ constants; $D_L(s) = 1 - B_{1,L}(s)$.
d, s	$d = L - 1$ the boundary regularity; d the constant shift in Section 12; s the exponential coordinate $z = \zeta e^{-s}$, and in Section 6.5 the regularity parameter of the hybrid theorem.
$F, \hat{F}, f_{K,y}$	F the shifted product (1.5); $\hat{F}$ without its first factor; $f_{K,y}$ the Newton function (9.17).
G, G_n	$G_n(z)$ the counting polynomial (2.1); $G(z)$ the subtracted function (6.7).
H	H_L the finite polylogarithmic exponent (3.1); H_m harmonic numbers ($H_0 = 0$).
$L, \mathcal{L}, M, m$	L the cutoff; $\mathcal{L}$ the Lambert target (9.13); $M = K + 2$ the number of derivatives; $m = n - 1$ the coefficient index.
P, Q	P_L, Q_L the factors of (3.2); $P_j(n, u), \tilde{P}_j$ the periodic polynomials; $Q_i = \hat{F}(i)$; $Q_{p,h,j}$ the transfer polynomials. <i>False reading</i> : P_j is not P_L .
r, q, h	r the polylogarithm index, the residue class in Proposition 5.1, and $r = r_K(y)$ the model root (9.9); q the order of a root; h a derivative order and a logarithm power.
T	$T(w)$ the tree function (4.3); $T_n = \sum k I_k$ in Section 2.3; $T(s), T_{d-1}$ Taylor polynomials in (3.4) and (6.4). <i>False reading</i> : T_n is a random variable.
t, ν, X	$t = 1 - z/\zeta$ the local variable, and t the Newton step count in Theorem 9.2; $\nu(y)$ the integer threshold; X the Lambert coordinate.
$\lambda_j, \omega, \varepsilon_\zeta$	$\lambda_j = \text{lcm}(2, \dots, j)$; $\omega = e^{2\pi i/3}$; $\varepsilon_\zeta = 1$ at $\pm i$.

[write] **Independent check of the write, 7 October 2026.** An independent adversarial check of the batch-113 write (f24d4b02b) used its own code. It read A368246 (#22), A143946 (#38) and A080130 again. The quotations of Remark 11.1 are verbatim, including Kotěšovec’s formula line.

Checked hardest, and confirmed: Kotěšovec’s conjecture, as posted, is proved.

- Theorem 2.1 of Giuliano–Szewczak–Weber was read in arXiv:1309.1578v1, which was fetched again and has the write’s SHA-256 value. It is stated on p. 2 for every integer sequence κ_n with $\kappa_n/n \rightarrow x > 0$, with the model of Section 2. The check verified exactly, for $n \leq 30$, that $n \Pr(T_n = n) = b_n$ from the product of the Bernoulli generating functions. With $\kappa_n = n$ and $\rho(1) = 1$ this gives $a_n \sim e^{-\gamma}(n-1)!$. Theorem 1.1 gives it as well.
- One exact integer evaluation of (2.1) to $N = 2000$ reproduces all 452 b-file terms. A brute force over all permutations with $n \leq 8$ confirms both statistics and the equality of their distributions.
- With exact b_n , the remainder of (1.4) times $n^4/(1 + \log n)^2$ is 0.650, -5.517 , -3.998 , 0.295 , -2.616 at $n = 100, 200, 500, 1000, 2000$, and at most 5.5171 in absolute value for $200 \leq n \leq 2000$, as the write found.

Also confirmed: C_ω from the Gamma product at 50 digits, with the note after Theorem 1.2 (real part a truncation, imaginary part a rounding); Q_i against extrapolated partial products; $S = 2 \log 2 + 1 - \zeta(2)$; and the constant-shift numbers. The transseries classification of Remark 9.3 was checked against the volume’s statements: $X(\log X - 1) = \mathcal{L}$ is `p0:thm:lambert-core` with $w = \log X - 1$, slope 1, coefficient 1. Also confirmed: the four `SOURCES.md` blobs (two current; the other two in 3412ae074 and a4198a037); the byte identity of the staged files; the README listing; and the numbering, with 104 delivered labels and 0 differences against the `.aux` of a build of the placed text.

Corrected or qualified by bracketed dated notes:

- the archive’s file count in the provenance paragraph above (24 files, not 25);
- the word “proved” for the constant-shift observation, in the note to Section 12.

2 Exact counting and the shifted product

2.1 The complete finite counting polynomial

For a permutation $\pi \in S_n$, write $c(\pi)$ for the sum of its cycle minima. Define

$$G_n(z) = \sum_{\pi \in S_n} z^{c(\pi)}, \quad G_0(z) = 1.$$

Proposition 2.1. *For every integer $n \geq 1$,*

$$G_n(z) = \prod_{k=1}^n (k - 1 + z^k). \quad (2.1)$$

The same polynomial enumerates the sum of record positions in permutations of $[n]$. Consequently $a_n = [z^n]G_n(z)$ for $n \geq 0$.

Proof. Insert the largest label n into a permutation of $[n-1]$ written in cycles. There are $n-1$ possible insertions immediately after one of the existing labels. Each leaves every cycle minimum unchanged. The remaining possibility creates the singleton cycle (n) , increasing the sum of minima by n . Every permutation of $[n]$ is obtained exactly once in this way. Thus

$$G_n(z) = (n - 1 + z^n)G_{n-1}(z),$$

and iteration proves (2.1).

For the second interpretation, let I_k indicate that position k is a left-to-right maximum of a uniform random permutation. The relative rank of the entry in position k among the first k entries is uniform on $\{1, \dots, k\}$, and these relative ranks are independent. One way to see independence is that the relative-rank code is a bijection between permutations and the $1 \cdot 2 \cdots n = n!$ possible codes. Therefore I_k are independent with $\Pr(I_k = 1) = 1/k$, and

$$\mathbb{E}_z \sum_{k=1}^n k I_k = \prod_{k=1}^n \left(1 - \frac{1}{k} + \frac{z^k}{k} \right) = \frac{G_n(z)}{n!}.$$

This proves equality in distribution of the two statistics and hence the equality of their counting polynomials. It is not a claim that the two sums agree for each individual permutation. $\square$

Factoring out the first factor z and the constants in the other factors gives

$$G_n(z) = z(n-1)! \prod_{j=1}^{n-1} \left(1 + \frac{z^{j+1}}{j} \right) \quad (n \geq 1). \quad (2.2)$$

Thus

$$b_n = [z^{n-1}] \prod_{j=1}^{n-1} \left(1 + \frac{z^{j+1}}{j} \right) = [z^{n-1}] F(z). \quad (2.3)$$

The factor with $j = n-1$ already has degree n and is harmless for this coefficient; the same is true of every subsequent factor. The equality is exact for $n = 1$ as well: the constant coefficient is 1. At $n = 2$ the coefficient of z vanishes, explaining the exceptional zero $a_2 = 0$.

2.2 Analytic existence and a global growth bound

For each $0 \leq r < 1$,

$$\sum_{j \geq 1} \frac{r^{j+1}}{j} = -r \log(1 - r) < \infty.$$

The product (1.5) therefore converges locally uniformly on $\mathbb{D} = \{z : |z| < 1\}$. It defines an analytic function with nonnegative Taylor coefficients. None of the factors vanishes in the disk, since its zeros have modulus $j^{1/(j+1)} \geq 1$. Hence F is nonzero in $\mathbb{D}$. Moreover,

$$|F(z)| \leq F(|z|) \leq \exp\left(\sum_{j \geq 1} \frac{|z|^{j+1}}{j}\right) = (1 - |z|)^{-|z|} \leq \frac{1}{1 - |z|}. \quad (2.4)$$

This is the finite global order -1 used in the classical hybrid theorem. It also controls the function independently of any local expansion.

2.3 The leading equivalent was already implied by a local limit theorem

Put $T_n = \sum_{k=1}^n kI_k$. Proposition 2.1 gives

$$\Pr(T_n = n) = \frac{a_n}{n!}, \quad n \Pr(T_n = n) = b_n.$$

Theorem 2.1 of [5] states that, for integer κ_n with $\kappa_n/n \rightarrow x > 0$,

$$n \Pr(T_n = \kappa_n) \longrightarrow e^{-x} \rho(x),$$

where the Dickman function satisfies $\rho(x) = 1$ on $[0, 1]$. Taking $\kappa_n = n$ gives $b_n \rightarrow A$. This report also computes A directly from the product, because that residue enters the global subtraction. Neither the alternative derivation nor its use here makes the leading equivalent a new result.

3 A uniformly differentiated tail on the closed disk

3.1 Why the first factor is separated

The logarithmic power series for $\log(1 + w)$ is well behaved uniformly for $|w| \leq 1/2$. In (1.5), this applies to every factor with $j \geq 2$ but not to $j = 1$. In fact $1 + z^2$ vanishes at $\pm i$. Hiding that factor in an infinite logarithmic tail would destroy the asserted arbitrary boundary smoothness precisely where it is needed.

For an integer $L \geq 1$, define

$$\begin{aligned} H_L(z) &= \sum_{r=1}^L \frac{(-1)^{r+1}}{r} z^r (\text{Li}_r(z^r) - z^r), \\ R_L(z) &= \sum_{r>L} \sum_{j \geq 2} \frac{(-1)^{r+1}}{r j^r} z^{r(j+1)}, \\ P_L(z) &= (1 + z^2) \exp H_L(z), \quad Q_L(z) = \exp R_L(z). \end{aligned} \quad (3.1)$$

Here $\text{Li}_r(w) = \sum_{j \geq 1} w^j / j^r$ for $|w| < 1$. Absolute convergence on compact subsets, or direct summation over $j \geq 2$, gives the exact factorization

$$F = P_L Q_L \quad \text{in } \mathbb{D}. \quad (3.2)$$

We use $C^d(\overline{\mathbb{D}})$ to mean that the analytic function and its complex derivatives through order d extend continuously to the closed disk. This entails C^d regularity of its boundary trace as a function of the angular variable. It does not entail analytic continuation outside the disk.

Lemma 3.1 (Uniform differentiated tail). *For $0 \leq d \leq L - 1$, the series defining R_L and all of its complex derivatives through order d converge absolutely and uniformly on $\overline{\mathbb{D}}$. In particular $Q_L, Q_L^{-1} \in C^d(\overline{\mathbb{D}})$, and their derivatives through order d are bounded there.*

Proof. For an integer $0 \leq h \leq d$, differentiating a summand at most h times gives, for $|z| \leq 1$,

$$\begin{aligned} \left| \frac{d^h}{dz^h} \frac{z^{r(j+1)}}{r j^r} \right| &\leq r^{h-1} \frac{(j+1)^h}{j^r} \\ &\leq \left(\frac{3}{2} \right)^h r^{h-1} j^{h-r}. \end{aligned}$$

The same inequality covers $h = 0$, with $r^{h-1} = r^{-1}$. Since $r \geq L + 1$ and $h \leq L - 1$, we have $u = r - h \geq 2$. Monotonicity and an integral comparison yield

$$\sum_{j \geq 2} j^{-u} \leq 2^{-u} + \int_2^\infty x^{-u} dx = 2^{-u} \left(1 + \frac{2}{u-1} \right) \leq 3 \cdot 2^{-u}.$$

Therefore the double sum of derivative bounds is at most

$$3 \left(\frac{3}{2} \right)^h \sum_{r > L} r^{h-1} 2^{h-r} < \infty. \quad (3.3)$$

The Weierstrass criterion supplies uniform absolute convergence, and termwise differentiation follows first in the disk and then by continuity at the boundary. Composition with the exponential gives the asserted regularity of Q_L and its reciprocal. Their boundedness follows from compactness; neither exponential can vanish. $\square$

3.2 Boundary Taylor expansion with a differentiated remainder

The following elementary point prevents a radial expansion from being mistaken for a boundary estimate.

Lemma 3.2. *Let $Q \in C^d(\overline{\mathbb{D}})$ with $d \geq 1$, and let $\zeta \in \partial\mathbb{D}$. If T_{d-1} is the Taylor polynomial of degree $d - 1$ formed from the boundary derivatives of Q at ζ , then*

$$(Q - T_{d-1})^{(h)}(z) = O(|z - \zeta|^{d-h}) \quad (0 \leq h \leq d), \quad z \in \overline{\mathbb{D}}. \quad (3.4)$$

The bound for $h = d$ means boundedness, not convergence to zero.

Proof. The closed disk is convex. For $h < d$, integrate $Q^{(d)}$ repeatedly along the line segment from ζ to z , obtaining

$$(Q - T_{d-1})^{(h)}(z) = \frac{(z - \zeta)^{d-h}}{(d-h-1)!} \int_0^1 (1-u)^{d-h-1} Q^{(d)}(\zeta + u(z - \zeta)) du.$$

The boundedness of $Q^{(d)}$ proves (3.4). For $h = d$ the derivative of T_{d-1} vanishes and the conclusion is immediate. Boundary segments follow by continuity from segments inside the disk. $\square$

Near a fixed ζ , use $z = \zeta e^{-s}$ with the branch satisfying $s = 0$ at $z = \zeta$. The change of variable is analytic with nonzero derivative. The bounds (3.4) remain valid with $|s|$ and s -derivatives, after changing constants. Inside the disk $\operatorname{Re} s \geq 0$; along a short unit-circle arc s is purely imaginary. Thus the same differentiated estimate controls both approaches.

3.3 What the finite factor contributes

Only the finite sum H_L can introduce a nonsmooth local contribution at the roots whose orders are at most L . Away from that finite set, every $\operatorname{Li}_r(z^r)$ in H_L continues analytically across the circle. Near a selected root, the integer polylogarithm expansion in Section 4 expresses it as an analytic function plus an analytic function times $\log s$. The first factor $1 + z^2$ is retained explicitly. Combined with Lemma 3.2, this gives finite power-log expansions with the differentiated remainders needed later. It does not replace the full product by a globally analytic finite product.

4 Rooted trees and cancellation at the positive singularity

4.1 The singular logarithm of an integer polylogarithm

For an integer $r \geq 1$, the local expansion on a fixed slit neighborhood is

$$\begin{aligned} \operatorname{Li}_r(e^{-rs}) &= \frac{(-rs)^{r-1}}{(r-1)!} (H_{r-1} - \log(rs)) \\ &\quad + \sum_{\substack{k \geq 0 \\ k \neq r-1}} \zeta(r-k) \frac{(-rs)^k}{k!}, \end{aligned} \tag{4.1}$$

where $H_0 = 0$ and $H_m = \sum_{h=1}^m h^{-1}$. For each of the finitely many $r \leq L$, the analytic part is convergent sufficiently near $s = 0$. With a fixed choice of $\log s$, (4.1) is valid on both sides of the short boundary arc under consideration. The singularity comes from the one displayed logarithm.

Multiplying by the coefficient $(-1)^{r+1} e^{-rs}/r$ from H_L gives the logarithmic part

$$-\frac{r^{r-1}}{r!} s^{r-1} e^{-rs} \log s.$$

Both sign factors $(-1)^{r+1}$ and $(-1)^{r-1}$ have canceled; the remaining minus sign is from $-\log(rs)$. Hence, at $z = e^{-s}$, the logarithmic coefficient of H_L is $-B_{1,L}(s)$, where

$$B_{1,L}(s) = \sum_{r=1}^L \frac{r^{r-1}}{r!} s^{r-1} e^{-rs}. \tag{4.2}$$

4.2 The exact tree identity

Let

$$T(w) = \sum_{r \geq 1} \frac{r^{r-1}}{r!} w^r. \quad (4.3)$$

Lagrange inversion gives $T(w) = we^{T(w)}$ for the analytic branch with $T(0) = 0$. Equivalently, $T(w)e^{-T(w)} = w$. Since $s \mapsto se^{-s}$ has derivative 1 at zero, the inverse function theorem implies

$$T(se^{-s}) = s \quad (4.4)$$

for s in a neighborhood of zero. Dividing its power series by s , with the removable value at $s = 0$, yields

$$\sum_{r \geq 1} \frac{r^{r-1}}{r!} s^{r-1} e^{-rs} = 1. \quad (4.5)$$

Every fixed Taylor coefficient receives contributions from finitely many r . Consequently

$$B_{1,L}(s) = 1 + O(s^L), \quad D_L(s) := 1 - B_{1,L}(s) = O(s^L). \quad (4.6)$$

No interchange of a nonconvergent boundary logarithmic series is used: the finite factorization is already exact, and (4.5) is an analytic identity near zero.

Proposition 4.1 (Only a pole remains at the positive root). *For every finite L , there is a function U_L analytic and nonzero near zero such that*

$$P_L(e^{-s}) = s^{-1} U_L(s) \exp(D_L(s) \log s). \quad (4.7)$$

The factor $Q_L(e^{-s})$ has the differentiated finite Taylor expansions of Lemma 3.2. It follows that, to every prescribed finite order, the only nonanalytic coefficient-producing term at $z = 1$ is a constant multiple of $(1 - z)^{-1}$.

Proof. The terms of $\log P_L(e^{-s})$ other than $-B_{1,L}(s) \log s$ are analytic near zero. This includes $\log(1 + e^{-2s})$, whose argument is nonzero there. Exponentiating the analytic part produces U_L . Since $-B_{1,L} = -1 + D_L$, equation (4.7) follows.

Now fix the desired finite order, then choose L larger. The logarithmic exponential differs from 1 first at order $s^L \log s$, so after multiplication by s^{-1} its first possible logarithm is at order $s^{L-1} \log s$. Before that order, multiplication by a sufficiently deep Taylor polynomial of Q_L produces a Laurent polynomial with one simple pole and otherwise nonnegative integer powers. The change of variable $s = -\log z$ is analytic and invertible near $z = 1$, and $s^{-1} - (1 - z)^{-1}$ is analytic there. Thus the sole singular contribution at the selected orders is the simple pole. $\square$

The conclusion is a finite-jet assertion for each requested differentiability order. It is not a convergent local Laurent expansion of F in a full neighborhood of 1. Dense singularities on the unit circle will rule out that stronger claim.

4.3 The pole coefficient from the product

For real $0 < z < 1$,

$$\begin{aligned} \log((1-z)F(z)) &= (1-z)\log(1-z) \\ &+ \sum_{j \geq 1} \left\{ \log\left(1 + \frac{z^{j+1}}{j}\right) - \frac{z^{j+1}}{j} \right\}. \end{aligned} \quad (4.8)$$

For $j \geq 2$ the summand is uniformly bounded in absolute value by a fixed multiple of j^{-2} ; the first summand is handled separately. Dominated convergence gives

$$\begin{aligned} \lim_{z \uparrow 1} \log((1-z)F(z)) &= \sum_{j \geq 1} (\log(1 + 1/j) - 1/j) \\ &= \lim_{N \rightarrow \infty} (\log(N+1) - H_N) = -\gamma. \end{aligned}$$

Therefore

$$\lim_{z \uparrow 1} (1-z)F(z) = e^{-\gamma} = A. \quad (4.9)$$

The finite-jet form identifies this radial limit with the coefficient of the local pole. This is a direct residue computation, rather than a Tauberian inference from the coefficient limit. Since $[z^m](1-z)^{-1} = 1$, the positive root contributes exactly A to every coefficient in the final subtraction.

5 Nontrivial roots and Gamma amplitudes

5.1 The root filtered tree series

Let $\zeta \neq 1$ have exact order $q \geq 2$, and put $z = \zeta e^{-s}$. In the term of index r of H_L , a polylogarithmic singularity occurs precisely when $\zeta^r = 1$, or equivalently $q \mid r$. For these terms, the same sign calculation as at the positive root applies. Define

$$B_{q,L}(s) = \sum_{\substack{1 \leq r \leq L \\ q \mid r}} \frac{r^{r-1}}{r!} s^{r-1} e^{-rs}. \quad (5.1)$$

The corresponding full analytic power series near zero can be expressed as

$$B_q(s) = \frac{1}{qs} \sum_{h=0}^{q-1} T(\exp(2\pi i h/q) s e^{-s}). \quad (5.2)$$

The apparent singularity at $s = 0$ is removable. Formula (5.2) is simply the root-of-unity filter applied to (4.3). Its first term is

$$B_q(s) = \frac{q^{q-2}}{(q-1)!} s^{q-1} + O(s^q). \quad (5.3)$$

At any fixed Taylor degree, $B_{q,L}$ agrees with B_q once L is large enough. Its omitted terms have degree at least L .

Let $\varepsilon_\zeta = 1$ if $\zeta = \pm i$ and 0 otherwise. For a sufficiently small slit neighborhood,

$$P_L(\zeta e^{-s}) = s^{\varepsilon_\zeta} U_{\zeta,L}(s) \exp(-B_{q,L}(s) \log s), \quad (5.4)$$

where $U_{\zeta,L}$ is analytic and nonzero at zero. The factor s^{ε_ζ} is precisely the isolated zero of $1 + z^2$; it is not absorbed into a branch of its logarithm. The other polylogarithms are analytic at this root, and their exponentials do not vanish.

5.2 Convergence of the boundary product

For $\zeta \neq 1$ on the unit circle, the series

$$\sum_{j \geq 1} \frac{\zeta^{j+1}}{j}$$

converges by Dirichlet's test. For $j \geq 2$, the difference between $\log(1 + \zeta^{j+1}/j)$ and its linear term is absolutely summable. The logarithm can be chosen from its convergent series because the argument has distance at least $1/2$ from zero. The first factor is dealt with separately. Thus the product has a finite nonzero limit except when its first factor vanishes, which happens at $\zeta = \pm i$.

The limit here means the original ordering of factors. At roots of unity, grouping finitely many consecutive residue classes preserves convergence. The same value is approached from inside the disk: the linear logarithmic term obeys Abel's theorem, and the quadratic remainder converges uniformly by its j^{-2} majorant. This also identifies the product limit with the constant term of the finite boundary jet.

Proposition 5.1 (Gamma product for a root value). *If ζ has exact order $q \geq 2$ and $\zeta \neq \pm i$, then*

$$F(\zeta) = \prod_{r=1}^q \frac{\Gamma(r/q)}{\Gamma((r + \zeta^{r+1})/q)} \neq 0. \quad (5.5)$$

Proof. Set $c_r = \zeta^{r+1}$. For the product over $j = q\ell + r$, $0 \leq \ell < N$, the Gamma recurrence gives

$$\begin{aligned} \prod_{\ell=0}^{N-1} \left(1 + \frac{c_r}{q\ell + r}\right) &= \prod_{\ell=0}^{N-1} \frac{\ell + (r + c_r)/q}{\ell + r/q} \\ &= \frac{\Gamma(N + (r + c_r)/q)}{\Gamma(N + r/q)} \frac{\Gamma(r/q)}{\Gamma((r + c_r)/q)}. \end{aligned}$$

The ratio depending on N is $N^{c_r/q}(1 + O(N^{-1}))$. Multiplication over $r = 1, \dots, q$ cancels all powers of N , since

$$\sum_{r=1}^q c_r = \sum_{r=1}^q \zeta^{r+1} = 0.$$

Taking $N \rightarrow \infty$ proves (5.5).

A factor of the original product can vanish only if $j = 1$ and $\zeta^2 = -1$. Equivalently, in the Gamma expression only $r = 1$, $c_1 = -1$ can make a denominator argument zero or a negative integer: for $r \geq 2$, its real part is at least $(r - 1)/q > 0$. The excluded roots are therefore the only zeros. This also follows directly from convergence of the logarithmic remainder. $\square$

5.3 The leading singularity and the natural boundary

For $\zeta \neq 1, \pm i$ of exact order q , let $t = 1 - z/\zeta$. We have $s = -\log(1 - t) = t + O(t^2)$. Equations (5.4) and (5.3), multiplied by the Taylor jet of Q_L , show that the first nonanalytic term is

$$-F(\zeta) \frac{q^{q-2}}{(q-1)!} t^{q-1} \log t. \quad (5.6)$$

Its coefficient is nonzero. Choosing L so that the smooth tail has more than q derivatives rules out cancellation of the logarithmic divergence by the remainder. In particular, F does not extend holomorphically through this root.

Corollary 5.2. *The unit circle is a natural boundary of F .*

Proof. Roots of unity other than 1 and $\pm i$ are dense on the circle, and each is singular by (5.6). Analytic continuation through any open arc would contradict one of those singularities. The radius of convergence is exactly 1, already ensured by the pole at 1. $\square$

For later reference, the generic leading coefficient from a single order- q root, expressed in the coefficient index m , is

$$(-1)^{q+1} q^{q-2} F(\zeta) \zeta^{-m} m^{-q}. \quad (5.7)$$

This follows from the transfer formula in Section 7. It is the leading contribution of that root, not the whole term of order m^{-q} , since roots of smaller order also have higher corrections.

5.4 The zeros at the order four roots

Define the product with its first factor removed by

$$\hat{F}(z) = \frac{F(z)}{1+z^2} = \prod_{j \geq 2} \left(1 + \frac{z^{j+1}}{j} \right).$$

The same convergence proof gives nonzero boundary values at $\pm i$. Grouping the factors at i gives

$$Q_i := \hat{F}(i) = \frac{\Gamma(5/4)\Gamma(1/2)\Gamma(3/4)}{\Gamma((2-i)/4)\Gamma((4+i)/4)} \neq 0. \quad (5.8)$$

For example, the residue class $j \equiv 1 \pmod{4}$ starts at $j = 5$ because $j = 1$ has been removed; its Gamma ratio is $\Gamma(5/4)/\Gamma(1)$. The other three residue classes give the remaining factors in (5.8).

At $z = ie^{-s}$,

$$1 + z^2 = 1 - e^{-2s} = 2s + O(s^2), \quad B_4(s) = \frac{8}{3}s^3 + O(s^4).$$

Hence the first nonanalytic term of F is

$$-\frac{16}{3} Q_i s^4 \log s = -\frac{16}{3} Q_i t^4 \log t + \text{higher nonanalytic terms}. \quad (5.9)$$

Thus the order-four roots first affect coefficients at order m^{-5} , rather than m^{-4} . Indeed their conjugate pair contributes at its first order

$$256 \operatorname{Re}(Q_i i^{-m}) m^{-5}. \quad (5.10)$$

This is only the contribution of this pair at that order. It is not a claimed complete formula for P_5 . The zero delays the logarithmic singularity; it does not remove it. In particular $\pm i$ are singular as well, although they were unnecessary for the density argument.

6 From local jets to a global coefficient remainder

The next argument is the analytic step that distinguishes a coefficient theorem from a radial heuristic. We fix a target order and verify the boundary regularity required for Fourier integration by parts.

6.1 Choice of finite cutoffs

Fix $K \geq 2$, and set

$$M = K + 2, \quad L = K + 6, \quad d = L - 1 = K + 5 = M + 3. \quad (6.1)$$

By Lemma 3.1, Q_L has bounded continuous complex derivatives through order d . Let Z_L be the finite set of roots of unity with exact order at most L . This is the singular set of the finite factor P_L on the unit circle. It is not the full singular set of F .

At each nontrivial $\zeta \in Z_L$, use (5.4). Expand its analytic factor in Taylor series, replace Q_L by its Taylor polynomial of degree $d - 1$, and expand the exponential of $-B_{q,L} \log s$. Retain every nonanalytic term that becomes

$$c_{\zeta,p,h} \left(1 - \frac{z}{\zeta}\right)^p \log^h \left(1 - \frac{z}{\zeta}\right), \quad 0 \leq p \leq M - 1, \quad h \geq 1, \quad (6.2)$$

after the analytic change from s to $t = 1 - z/\zeta$. There are only finitely many such terms. The minimum degree in $B_{q,L}$ is $q - 1 \geq 1$, so a power $(\log s)^h$ from the exponential has degree at least $h(q - 1)$ before the possible extra zero. This ensures finiteness without assigning a cutoff in logarithm degree separately.

The coefficients $c_{\zeta,p,h}$ are defined by this finite-jet operation. They do not depend on increasing L once sufficient differentiability and Taylor depth are available: two valid expansions with controlled higher-order remainders have identical lower-order nonanalytic jets. Their values can include convergent derivatives of Q_L , rather than only elementary constants. The Gamma products are their leading amplitudes.

6.2 Remainders at the nontrivial selected roots

We record why each step in this local construction remains valid after M derivatives. For a fixed integer $p \geq M$ and $h \geq 0$,

$$\frac{d^M}{ds^M} (s^p \log^h s) = O(|s|^{p-M} (1 + |\log s|)^h) \quad (6.3)$$

on the relevant punctured neighborhood. Derivatives of lower orders have the analogous extra positive powers. In particular, when $p = M$, the M th derivative is locally integrable on a boundary arc, and derivatives through order $M - 1$ tend to zero for this remainder.

The exponential in (5.4) is expanded to enough terms that its first omitted power has s -degree at least M . Its usual Taylor formula, applied to the small quantity $B_{q,L}(s) \log s$, yields the same differentiated bounds. Truncating analytic factors is equally harmless after choosing their Taylor depth beyond M .

The only nonanalytic object not represented by a convergent power-log expansion is Q_L . Lemma 3.2 gives, in the s coordinate,

$$(Q_L(\zeta e^{-s}) - T(s))^{(h)} = O(|s|^{d-h}) \quad (0 \leq h \leq d), \quad (6.4)$$

for a Taylor polynomial T of degree $d - 1$. Derivatives of P_L of order $v \leq M$ consist of bounded analytic contributions and terms bounded by

$$O(|s|^{\varepsilon_\zeta + q - 1 - v} (1 + |\log s|)^{c_v})$$

for fixed finite exponents c_v . The Leibniz rule applied to $P_L(Q_L - T)$ now gives positive powers of $|s|$ after M derivatives. For the singular part the exponent is at least

$$d + \varepsilon_\zeta + q - 1 - M \geq 4,$$

and the bounded analytic part gives at least $d - M = 3$. All these terms are locally integrable, with lower derivatives continuous at the root. This explicitly checks the tail remainder rather than treating it as an analytic function across the boundary.

The analytic change $s = -\log(1 - t)$ preserves these bounds: s/t is analytic and nonzero near zero, and

$$\log s = \log t + \log(s/t)$$

adds only an analytic factor. Differentiation along the boundary angular coordinate also preserves the local integrability and continuity properties, since it is a nonsingular smooth coordinate change.

6.3 The remainder at the positive root

At 1, equation (4.7) gives

$$F(e^{-s}) = s^{-1}U_L(s)Q_L(e^{-s})\exp(D_L(s)\log s), \quad D_L(s) = O(s^L).$$

The first logarithmic term, including the pole, has degree $L - 1$. Its M th derivative is bounded by

$$O(|s|^{L-1-M}(1 + |\log s|)^c) = O(|s|^3(1 + |\log s|)^c). \quad (6.5)$$

The remainder from replacing Q_L by its degree- $(d - 1)$ Taylor polynomial and dividing by s has M th derivative

$$O(|s|^{d-M-1}) = O(|s|^2). \quad (6.6)$$

After the pole $A/(1 - z)$ is subtracted, everything else up to these remainders is analytic near 1. Equations (6.5)–(6.6) therefore give the same required boundary regularity there.

6.4 A single analytic subtraction in the disk

For each $\zeta \in \partial\mathbb{D}$, the function $\log(1 - z/\zeta)$ has its usual analytic branch in $\mathbb{D}$, since $1 - z/\zeta$ has positive real part there. Let J_ζ be the finite sum (6.2) at the nontrivial root ζ . Define

$$G(z) = F(z) - \frac{A}{1 - z} - \sum_{\zeta \in Z_L \setminus \{1\}} J_\zeta(z). \quad (6.7)$$

Every subtraction is analytic in the open disk. At a different selected root, a fixed J_ζ is locally analytic. Subtracting it may change ordinary Taylor coefficients but cannot change a nonanalytic jet at that other root. Thus all of the local estimates above hold simultaneously for G .

Away from the finite set Z_L , the finite factor P_L continues analytically and Q_L has d bounded boundary derivatives. The high-order roots not in Z_L are included in that smooth tail. They do not create an unexamined exceptional set in the argument. A finite covering of the circle by selected-root neighborhoods and their complement gives a global conclusion.

Lemma 6.1 (Boundary regularity and Fourier estimate). *The function G in (6.7) extends continuously to $\overline{\mathbb{D}}$. Its boundary trace $g(\theta) = G(e^{i\theta})$ has continuous derivatives through order $M - 1$, these derivatives are absolutely continuous, and $g^{(M)} \in L^1[0, 2\pi]$. Consequently*

$$[z^m]G(z) = O_K(m^{-M}) = O_K(m^{-K-2}). \quad (6.8)$$

Proof. The local remainders have continuous derivatives through order $M - 1$ by the estimates just established. On each punctured arc they possess an ordinary M th derivative that is integrable, since all powers of $|\log|\theta - \theta_0||$ are locally integrable. The lower-order derivatives have matching limits from either side. The fundamental theorem of calculus on truncated arcs, followed by the integrable limit, proves absolute continuity across the removed point. A finite partition then proves global absolute continuity. This also verifies the asserted regularity of all lower derivatives.

Since G is continuous on the compact closed disk, its radial boundary values converge uniformly. Cauchy's coefficient formula therefore identifies its Taylor coefficients with the Fourier coefficients of g . Repeated integration by parts, with periodic boundary terms canceling, gives for $m \geq 1$

$$[z^m]G(z) = \frac{(im)^{-M}}{2\pi} \int_0^{2\pi} g^{(M)}(\theta) e^{-im\theta} d\theta. \quad (6.9)$$

The L^1 norm proves (6.8). The Riemann–Lebesgue lemma would strengthen this to a little-oh estimate at the same power, but is not needed. $\square$

6.5 Relationship to the published hybrid theorem

This is a direct boundary proof within the classical Darboux and hybrid framework [7]. To invoke their Theorem 2 literally instead, one can choose the larger cutoff $L = 2K + 6$ and boundary regularity $s = 2K + 5$. The singular factor has global order $a = -1$, because (2.4) holds and Q_L^{-1} is bounded. Their exponent is

$$u_0 = \left\lfloor \frac{s + \lfloor a \rfloor}{2} \right\rfloor = K + 2.$$

The finite polylogarithms continue into a common indented domain at the finitely many roots and have arbitrarily deep power-log expansions, so the remaining hypotheses can be met by choosing their local expansion depth $t_0 > u_0$. Their conclusion gives the required coefficient remainder. The smaller cutoff $L = K + 6$ in (6.1) is justified by the direct derivative argument above; it is not obtained by silently substituting into the published exponent formula.

7 Coefficient transfer and the zero mean sector

7.1 An exact identity before asymptotic expansion

For $m \geq 0$ and a complex parameter α ,

$$[z^m] \left(1 - \frac{z}{\zeta}\right)^\alpha = \zeta^{-m} (-1)^m \binom{\alpha}{m} = \zeta^{-m} \frac{\Gamma(m - \alpha)}{\Gamma(-\alpha)\Gamma(m + 1)}. \quad (7.1)$$

The binomial coefficient is a polynomial in α , so the Gamma expression is interpreted by analytic continuation at apparent poles. Differentiate h times with respect to α and then set $\alpha = p \in \mathbb{Z}_{\geq 0}$ to obtain the exact coefficient of

$$\left(1 - \frac{z}{\zeta}\right)^p \log^h \left(1 - \frac{z}{\zeta}\right).$$

For a fixed finite set of p, h , the Gamma-ratio expansion may be differentiated uniformly in a small parameter neighborhood. This follows, for example, by applying Cauchy's derivative formula to the uniform finite Stirling expansion in that neighborhood. Thus it retains a controlled remainder after differentiation.

The ratio in (7.1) begins with $m^{-\alpha-1}$. At a nonnegative integer p , the reciprocal Gamma factor $1/\Gamma(-\alpha)$ has a simple zero. Consequently the coefficient at every inverse-power order is a polynomial in $\log m$ of degree at most $h-1$, rather than h . More explicitly, for each fixed $J \geq p+1$,

$$[z^m] \left(1 - \frac{z}{\zeta}\right)^p \log^h \left(1 - \frac{z}{\zeta}\right) = \zeta^{-m} \sum_{j=p+1}^J \frac{Q_{p,h,j}(\log m)}{m^j} + O\left(\frac{(1 + \log m)^{h-1}}{m^{J+1}}\right), \quad (7.2)$$

where $h \geq 1$ and $\deg Q_{p,h,j} \leq h-1$. For $h=1$, one also has the particularly useful exact formula

$$[z^m] \left(1 - \frac{z}{\zeta}\right)^p \log \left(1 - \frac{z}{\zeta}\right) = \zeta^{-m} \frac{(-1)^{p+1} p!}{m(m-1) \cdots (m-p)}, \quad m > p. \quad (7.3)$$

This formula fixes the signs in both (5.7) and the low-order expansion.

7.2 Which roots can contribute at a given order

At a root of order $q \geq 2$, a logarithm power h in the expansion of (5.4) has degree at least $h(q-1)$, and the extra zero at $\pm i$ adds one more. Multiplication by analytic jets and change of analytic coordinate can only increase the degree. Thus every nonanalytic monomial has

$$p \geq h(q-1) + \varepsilon_\zeta. \quad (7.4)$$

If it contributes at order m^{-j} , then $j \geq p+1$. In particular,

$$q \leq j, \quad h-1 \leq p-1 \leq j-2. \quad (7.5)$$

This proves the bound on logarithm degree and shows that only roots of order at most j can appear at order j . The order-four exception improves the first restriction for that specific pair. There is no monomial with $p=0$ at a nontrivial root, hence no m^{-1} term.

7.3 Discarding the deeper subtraction terms

The global construction retained all nonanalytic terms through $p = M-1 = K+1$ so that its remainder had M integrable derivatives. To state an expansion only through m^{-K} , retain in the coefficient formula only the terms with $p \leq K-1$ and only the transfer powers $j \leq K$.

Terms with $p = K$ start at

$$O(m^{-K-1}(1 + \log m)^{K-1}).$$

Terms with $p = K+1$ start at $O(m^{-K-2}(1 + \log m)^K)$ and are absorbed in the same bound for large m . Terms from expanding the transfer of lower-degree monomials beyond $j = K$ also obey that bound. Finally, (6.8) is stronger. There are finitely many roots and monomials for each K , so addition gives

$$[z^m]F(z) = A + \sum_{j=2}^K \frac{\tilde{P}_j(m, \log m)}{m^j} + O_K\left(\frac{(1 + \log m)^{K-1}}{m^{K+1}}\right), \quad (7.6)$$

with the restrictions (7.5).

This two-depth argument is useful: one subtracts more local terms than one finally displays. Stopping the local analysis at the displayed order alone would not automatically furnish the global Fourier remainder.

7.4 Periodicity and the absence of a constant frequency

Every term in $\tilde{P}_j$ has a nontrivial frequency ζ^{-m} with exact order $q \leq j$. Its coefficient is a polynomial in $\log m$. The root 1 contributed only the constant A because of Section 4; it supplies no algebraic correction.

For any nontrivial ζ with $\zeta^{\lambda_j} = 1$,

$$\sum_{m=0}^{\lambda_j-1} \zeta^{-m} = 0. \quad (7.7)$$

Hence every logarithmic coefficient in $\tilde{P}_j$ is periodic with period dividing λ_j and has zero mean. Since F has real Taylor coefficients, the amplitudes at conjugate roots are conjugate. Pairing them makes the coefficients visibly real.

To finish, use the exact shift $m = n - 1$. For fixed j ,

$$(n-1)^{-j} = n^{-j}(1 - 1/n)^{-j}, \quad \log(n-1) = \log n + \log(1 - 1/n), \quad \zeta^{-m} = \zeta^{1-n}.$$

Taylor expansion to the requested finite order preserves the logarithm-degree bound. Contributions to order n^{-j} come from original orders at most j , whose periods all divide λ_j . Their frequencies stay nontrivial, and multiplication by the fixed factor ζ does not create a constant frequency. Equation (7.7) remains valid. The remainder transforms to that in (1.2), proving Theorem 1.1.

7.5 A finite prescription for every coefficient

The proof is constructive at a fixed order, although it does not promise elementary closed forms for all higher amplitudes. A precise prescription is:

1. Choose $L = K + 6$ and enumerate the roots in Z_L .
2. At each root, calculate the finite polylogarithmic analytic and logarithmic jets from (3.1), (4.1), and (5.4).
3. Obtain the required derivatives of Q_L from its absolutely and uniformly differentiated series. Formula (3.3) supplies convergent bounds for these derivatives.
4. Multiply jets and change to $t = 1 - z/\zeta$. Keep the nonanalytic monomials required by the two-depth subtraction.
5. Apply (7.1), expand the Gamma ratio to the chosen finite order, pair conjugate roots, and replace m by $n - 1$.

This specifies actual coefficients of this sequence. Generic power-log transfer examples are useful tests of the method, but their arbitrary input amplitudes are not higher coefficients of A368246. The accompanying code distinguishes the exact sequence calculations from symbolic transfer identities and only evaluates the fully displayed sequence models of orders two and three.

8 The second and third order terms

8.1 The value and analytic slope at minus one

At $z = -1$, consecutive product factors telescope:

$$\left(1 + \frac{1}{2k-1}\right) \left(1 - \frac{1}{2k}\right) = 1. \quad (8.1)$$

Thus $F(-1) = 1$. The first terms of the filtered tree series are

$$B_2(s) = s - 2s^2 + O(s^3). \quad (8.2)$$

To obtain the order-three coefficient, the analytic linear term of $\log F(-e^{-s})$ is also required. We compute it directly rather than infer it from finite coefficient data.

Isolate $j = 1$, as before, and group the logarithmic sum by the power r . The analytic coefficient of s in the $r = 1$ term is

$$\frac{3}{2} - \log 2,$$

and in the $r = 2$ term it is

$$\zeta(2) - 1 - \log 2.$$

For clarity, the former comes from differentiating $z(\text{Li}_1(z) - z)$ at $z = -e^{-s}$. For the latter, insert the expansion of $\text{Li}_2(e^{-2s})$ into $-\frac{1}{2}e^{-2s}(\text{Li}_2(e^{-2s}) - e^{-2s})$ and separate its $-s \log s$ part. The isolated factor $\log(1 + e^{-2s})$ contributes -1 .

For all $r \geq 3$, the first derivative is absolutely summable after $j = 1$ has been removed. Summing the geometric series in r gives the remaining analytic slope

$$\begin{aligned} S &= \sum_{j \geq 2} (j+1) \frac{((-1)^j/j)^3}{1 - (-1)^j/j} \\ &= \sum_{\substack{j \geq 2 \\ j \text{ even}}} \left(\frac{2}{j-1} - \frac{2}{j} - \frac{1}{j^2} \right) - \sum_{\substack{j \geq 3 \\ j \text{ odd}}} \frac{1}{j^2} \\ &= 2 \log 2 + 1 - \zeta(2). \end{aligned} \quad (8.3)$$

Adding the four contributions gives $1/2$. Therefore the local logarithm is

$$\begin{aligned} \log F(-e^{-s}) &= -s \log s + \frac{s}{2} + 2s^2 \log s \\ &\quad + \text{analytic terms of degree at least } 2 + O(s^3(1 + |\log s|)), \end{aligned} \quad (8.4)$$

where the finite-jet statement and its differentiated control follow from the earlier factorization. The omitted analytic coefficient of s^2 will not affect the nonanalytic terms of degree at most two.

8.2 Changing from the exponential coordinate

Set $t = 1 + z$, so $s = -\log(1 - t) = t + t^2/2 + O(t^3)$. Exponentiating (8.4) first gives the nonanalytic terms

$$-s \log s + s^2 \left(\frac{1}{2} \log^2 s + \frac{3}{2} \log s \right).$$

The analytic $s/2$ term matters through its product with $-s \log s$. On changing coordinates, the contribution of $-s \log s$ includes $-\frac{1}{2}t^2 \log t$. The final nonanalytic jet through degree two is therefore

$$-t \log t + t^2 \left(\frac{1}{2} \log^2 t + \log t \right). \quad (8.5)$$

This calculation explains why omitting the analytic linear coefficient, or forgetting the coordinate change, would alter the constant at order three.

8.3 Exact transfer at minus one

For $m \geq 3$, the basic identities are

$$\begin{aligned} [z^m](1-z) \log(1-z) &= \frac{1}{m(m-1)}, \\ [z^m](1-z)^2 \log(1-z) &= -\frac{2}{m(m-1)(m-2)}, \\ [z^m](1-z)^2 \log^2(1-z) &= \frac{4(H_{m-3} - 3/2)}{m(m-1)(m-2)}. \end{aligned} \quad (8.6)$$

The last identity follows either by twice differentiating (7.1) at $p = 2$, or by multiplying the identity $[z^m] \log^2(1-z) = 2H_{m-1}/m$ by $(1-z)^2$. Substituting $z \mapsto -z$ adds the factor $(-1)^m$.

Since $H_{m-3} = \log m + \gamma + O(m^{-1})$, the coefficient of (8.5) is

$$(-1)^m \left\{ -\frac{1}{m^2} + \frac{2 \log m + 2\gamma - 6}{m^3} \right\} + O\left(\frac{(1 + \log m)^2}{m^4} \right). \quad (8.7)$$

For example, the three contributions to the constant at m^{-3} are -1 from $-t \log t$, -3 from half the squared-log term, and -2 from $t^2 \log t$. Their sum is -6 .

Now $m = n - 1$ and $(-1)^m = -(-1)^n$. Expanding the denominators and logarithm gives

$$(-1)^n \left\{ \frac{1}{n^2} + \frac{8 - 2\gamma - 2 \log n}{n^3} \right\} + O\left(\frac{(1 + \log n)^2}{n^4} \right). \quad (8.8)$$

The change from 6 to 8 is the $m = n - 1$ shift of the leading term; this is not an optional convention.

8.4 The two cubic roots

For $\zeta = \omega$ or $\bar{\omega}$, equation (5.6) becomes

$$-\frac{3}{2}F(\zeta) \left(1 - \frac{z}{\zeta} \right)^2 \log \left(1 - \frac{z}{\zeta} \right).$$

By (7.3), its leading coefficient is

$$3F(\zeta)\zeta^{-m}m^{-3}.$$

The conjugate pair therefore contributes

$$\frac{6 \operatorname{Re}(C_\omega \omega^{-m})}{m^3} + O\left(\frac{1 + \log m}{m^4} \right). \quad (8.9)$$

Replacing m by $n - 1$ changes the phase to ω^{1-n} and does not otherwise change this leading order. No roots of order at least four contribute through order three. Combining the pole coefficient A , (8.8), and (8.9) proves Theorem 1.2.

8.5 The first periodic polynomials and numerical meaning

The first two polynomials in Theorem 1.1 are explicitly

$$\begin{aligned} P_2(n, u) &= (-1)^n, \\ P_3(n, u) &= (-1)^n(8 - 2\gamma - 2u) + 6 \operatorname{Re}(C_\omega \omega^{1-n}). \end{aligned}$$

Their periods divide 2 and 6, respectively. The coefficient of u and the constant coefficient in P_3 both average to zero over six consecutive integers. The absence of a smooth correction is visible at these first orders and is proved at every fixed order by the positive-root cancellation.

The code evaluates C_ω from the Gamma product and independently from the polylogarithmic logarithm with $j = 1$ removed. If the sum in that logarithm is truncated after $r = R$, its omitted absolute logarithmic tail is bounded by

$$\sum_{r>R} \frac{1}{r} \sum_{j \geq 2} j^{-r} \leq \frac{3 \cdot 2^{-R}}{R+1}. \quad (8.10)$$

This is an analytic truncation bound. Ordinary high precision evaluation of the retained transcendental terms still has numerical roundoff; the agreement of two evaluations is a diagnostic rather than a certified enclosure unless all those operations are evaluated with directed interval control. No coefficient in (1.4) is obtained by a numerical fit.

9 A real model and a genuine integer inverse

9.1 Fixing a real interpolation explicitly

An integer asymptotic formula does not determine a unique smooth interpolation. For inverse statements we therefore choose one, using only the finite expansion proved above. First collect each logarithmic coefficient into its root-of-unity Fourier representation. Replace the self-conjugate frequency $(-1)^n$ by $\cos(\pi x)$. For a nonreal conjugate pair, choose the member ζ with $0 < \arg \zeta < \pi$, and replace the integer pair by

$$2 \operatorname{Re} \left(c(\log x) e^{-ix \arg \zeta} \right). \quad (9.1)$$

The coefficient c includes any fixed phase resulting from the shift $m = n - 1$. This prescription is real for real x and agrees with the expansion at every integer. Using $e^{-\pi i x}$ alone for the frequency $(-1)^n$ would be incorrect for this purpose, because it is not real between integers.

Let $B_K(x)$ be the interpolation of the right-hand side of (1.2) with its error omitted, and put

$$A_K(x) = \Gamma(x) B_K(x). \quad (9.2)$$

For instance,

$$\begin{aligned} B_2(x) &= A + \frac{\cos(\pi x)}{x^2}, \\ B_3(x) &= A + \frac{\cos(\pi x)}{x^2} + \frac{\cos(\pi x)(8 - 2\gamma - 2 \log x)}{x^3} \\ &\quad + \frac{6 \operatorname{Re}(C_\omega \omega e^{-2\pi i x/3})}{x^3}. \end{aligned} \quad (9.3)$$

All following assertions refer to this fixed interpolation, rather than to an unknown smooth inverse of the actual integer sequence.

For each fixed K and each fixed derivative order h , termwise differentiation of the finite trigonometric expression gives

$$B_K(x) = A + O_K(x^{-2}), \quad B_K^{(h)}(x) = O_{K,h}(x^{-2}) \quad (h \geq 1). \quad (9.4)$$

Indeed, the leading term is of order x^{-2} and differentiation of its frequency does not increase its magnitude; the finitely many higher terms $x^{-j}(\log x)^{j-2}$ are smaller. Thus B_K is eventually positive. By the standard Stirling expansions of the digamma and trigamma functions,

$$\begin{aligned} \frac{d}{dx} \log A_K(x) &= \log x + O_K(x^{-1}) > 0, \\ \frac{d^2}{dx^2} \log A_K(x) &= \frac{1}{x} + O_K(x^{-2}) > 0. \end{aligned} \quad (9.5)$$

Both inequalities hold only for sufficiently large x , with an onset not evaluated here.

9.2 Eventual monotonicity of the integer sequence

The leading equivalent $b_n \rightarrow A > 0$ gives

$$\frac{a_{n+1}}{a_n} = n \frac{b_{n+1}}{b_n} \sim n. \quad (9.6)$$

Hence a_n is positive and strictly increasing for all sufficiently large n . Choose any fixed n_0 beyond that point. For sufficiently large y , define

$$\nu(y) = \min\{n \geq n_0 : a_n \geq y\}. \quad (9.7)$$

Changing the chosen n_0 has no effect when y is large. The low-index zero at $n = 2$ therefore creates no ambiguity in the eventual inverse.

At integer n , positivity and (1.2) imply

$$\log a_n - \log A_K(n) = O_K\left(n^{-K-1}(1 + \log n)^{K-1}\right). \quad (9.8)$$

This compares the model and the sequence only at integers. There is no assertion that a continuous interpolation of the sequence has the same error between integers.

9.3 The two ceiling envelope

For sufficiently large y , let $r = r_K(y)$ be the unique sufficiently large real solution of

$$A_K(r) = y. \quad (9.9)$$

Existence and uniqueness follow from eventual monotonicity of the model, its positivity, and $A_K(x) \rightarrow \infty$.

Theorem 9.1 (Integer inverse envelope). *For each fixed $K \geq 2$ there are constants $C_K > 0$ and y_K such that, with*

$$\delta_K(y) = C_K r_K(y)^{-K-1} (1 + \log r_K(y))^{K-2}, \quad (9.10)$$

one has

$$\lceil r_K(y) - \delta_K(y) \rceil \leq \nu(y) \leq \lceil r_K(y) + \delta_K(y) \rceil \quad (y \geq y_K). \quad (9.11)$$

For sufficiently large y , the two bounds are either identical or consecutive integers. In the latter case, comparison with the actual sequence at the lower candidate determines the answer.

Proof. On $[r - 2, r + 2]$, for sufficiently large r , (9.5) gives

$$\frac{d}{dx} \log A_K(x) \geq \frac{1}{2} \log r.$$

On the integers of this interval, the absolute error in (9.8) is at most

$$D_K r^{-K-1} (1 + \log r)^{K-1}$$

for a fixed D_K , after absorbing the bounded ratios between n and r . Choose C_K sufficiently larger than $2D_K$, including the harmless eventual ratio $(1 + \log r)/\log r$. Because $\delta_K \rightarrow 0$, the integers

$$j = \lceil r - \delta_K \rceil - 1, \quad m = \lceil r + \delta_K \rceil \quad (9.12)$$

belong to $[r - 2, r + 2]$ for large r . Their distances from r are at least δ_K with the indicated signs. The mean value theorem and the strictly enlarged constant imply

$$\log a_j < \log y, \quad \log a_m \geq \log y.$$

The strict lower index in (9.12) covers equality of y with a sequence value. Eventual monotonicity of a_n extends these comparisons to all smaller and larger indices, proving (9.11).

For large y , $2\delta_K < 1$, so the two ceilings differ by at most one. If the bounds coincide, that integer is $\nu(y)$. Otherwise let ℓ be the lower candidate. The answer is ℓ when $a_\ell \geq y$, and $\ell + 1$ when $a_\ell < y$. $\square$

The factor $(1 + \log r)^{K-2}$ is the counting error transported through a derivative of size $\log r$. It is one logarithmic power smaller than the error in (9.8). The envelope is an asymptotic theorem; the constants C_K, y_K have not been numerically certified, so it is not an unconditional finite rounding rule.

9.4 A Lambert W starting coordinate

Define

$$\mathcal{L} = \log y + \gamma - \frac{1}{2} \log(2\pi), \quad X = \frac{\mathcal{L}}{W_0(\mathcal{L}/e)}, \quad (9.13)$$

where W_0 is the principal real branch of Lambert's function. For sufficiently large y , $\mathcal{L} > 0$ and

$$X(\log X - 1) = \mathcal{L}. \quad (9.14)$$

Stirling's formula and (9.4) give, uniformly for bounded u ,

$$\log A_K(X + u) - \log y = (u - \tfrac{1}{2}) \log X + O_K(X^{-1}).$$

In particular the root lies in a fixed bounded neighborhood of $X + 1/2$. Evaluating more precisely at that center yields

$$\log A_K(X + \tfrac{1}{2}) - \log y = -\frac{1}{24X} + O_K(X^{-2}). \quad (9.15)$$

To check the coefficient, expand

$$\log \Gamma(X + \tfrac{1}{2}) = X \log(X + \tfrac{1}{2}) - X - \tfrac{1}{2} + \tfrac{1}{2} \log(2\pi) + \frac{1}{12(X + 1/2)} + O(X^{-3}).$$

The first two terms contribute $X \log X - X - 1/(8X) + O(X^{-2})$, and the Stirling correction adds $1/(12X)$, leaving $-1/(24X)$. The term $\log B_K = -\gamma + O_K(X^{-2})$ does not change this coefficient. By (9.5),

$$r_K(y) = X + \frac{1}{2} + O_K\left(\frac{1}{X \log X}\right). \quad (9.16)$$

This establishes the displacement from the Lambert coordinate without using a fitted inverse.

9.5 A fixed number of Newton steps reaches every fixed order

Set

$$f_{K,y}(x) = \log \Gamma(x) + \log B_K(x) - \log y, \quad (9.17)$$

and define the finite iterates

$$x_0 = X + 1, \quad x_{t+1} = x_t - \frac{f_{K,y}(x_t)}{f'_{K,y}(x_t)}. \quad (9.18)$$

Only the finite, explicitly chosen model B_K is used here.

Theorem 9.2 (Finite Newton transport). *For each fixed K , the iterates in (9.18) are well defined for all sufficiently large y , stay above $r_K(y)$, and decrease toward it. For every fixed integer $t \geq 0$,*

$$0 \leq x_t - r_K(y) = O_{K,t}\left((X \log X)^{-(2^t-1)}\right). \quad (9.19)$$

In particular, for

$$t_K = \lceil \log_2(K + 2) \rceil, \quad (9.20)$$

there exist constants $\tilde{C}_K, \tilde{y}_K$ such that

$$\lceil x_{t_K} - \Delta_K \rceil \leq \nu(y) \leq \lceil x_{t_K} + \Delta_K \rceil, \quad \Delta_K = \tilde{C}_K X^{-K-1} (1 + \log X)^{K-2}, \quad (9.21)$$

for $y \geq \tilde{y}_K$. Thus the center of the inverse envelope can be obtained by a finite number of specified operations, rather than left as an unspecified smooth inverse.

Proof. By (9.16), $x_0 = X + 1$ lies above the model root and differs from it by $O(1)$. On the bounded interval between the root and x_0 , equations (9.5) give

$$f'_{K,y}(x) \asymp \log X, \quad 0 < f''_{K,y}(x) = O_K(X^{-1}).$$

For an increasing convex function, the Newton intercept from a point to the right of a zero lies between that zero and the starting point. Therefore every iterate stays in this interval. Taylor's theorem, applied between r and x_t , gives

$$x_{t+1} - r = \frac{f''_{K,y}(\xi_t)}{2f'_{K,y}(x_t)}(x_t - r)^2 \leq \frac{C'_K}{X \log X}(x_t - r)^2 \quad (9.22)$$

for an intermediate point ξ_t . Starting from an $O(1)$ error and inducting proves (9.19), since the exponents obey $e_{t+1} = 2e_t + 1$, $e_0 = 0$.

For the choice (9.20), $2^{t_K} - 1 \geq K + 1$, so the error in (9.19) is bounded by a constant times $X^{-K-1}(1 + \log X)^{K-2}$. The quantities X and r are asymptotic by (9.16). Enlarging the constant to cover both the Newton error and the radius in Theorem 9.1 proves (9.21). $\square$

The theorem concerns exact mathematical iteration. A floating point implementation adds numerical error. The supplied evaluator for $K = 2, 3$ reports numerical iterates and residuals at a chosen precision; it does not certify interval enclosure of r_K , provide the unknown onset in (9.21), or license unconditional rounding. When the two integer candidates differ, a genuine threshold decision still uses the exact counting polynomial or another independently certified comparison with a_n .

Remark 9.3 ([write] The inverse and the transseries volume, 7 October 2026). Compared statement by statement with the repository's transseries volume [8]; the volume's symbols are not used here.

1. *The growth.* $\log a_n = \log \Gamma(n) + \log b_n = n \log n - n - \frac{1}{2} \log n + O(1)$ contains the term $n \log n$, so the sequence is not of the exponential–power form of `p0:def:model`; no instance of the volume's model is claimed.
2. *The Lambert coordinate.* With $w = \log X - 1$, the equation (9.14) becomes $w + \log w = \log \mathcal{L} - 1$: the volume's dominant block `p0:thm:lambert-core` with unit slope, unit logarithmic coefficient and target $\log \mathcal{L} - 1$, whose principal branch gives $w = W_0(\mathcal{L}/e)$, that is the displayed X . An exact instance.
3. *The model root and the Newton iterates.* The equation $\log A_K(x) = \log y$ differs from the Lambert block by Stirling terms and by the oscillatory terms of B_K , which in the chart of $\log x$ are exponentially small and are not series in its reciprocal with coefficients polynomial in its logarithm; the displacement (9.16) and the iterates of Theorem 9.2 are relative corrections of order $1/X$. So the write has not shown them to be instances of `plt:thm:lw-template`; they are proved directly.
4. *The thresholds.* $\nu(y)$ is the volume's staircase (`p0:def:three-inverses(1)`) of the eventually increasing sequence a_n . The model A_K interpolates the truncated expansion, not the sequence ($A_K(n) \neq a_n$, as Section 9 itself stresses), so it is not an admissible interpolation; Theorems 9.1 and 9.2, proved directly at the integers, are analogues of `p0:thm:staircase(2)`, not instances, and their final comparison with a_ℓ when the two ceilings differ is the volume's separation caveat.

10 Reproducible computations and their limits

The analytic theorem rests on the preceding proofs. The accompanying programs serve three narrower purposes: checking exact finite identities, checking the displayed symbolic calculations, and diagnosing numerical behavior without turning a finite fit into an asymptotic proof. All supplied code is editable and runs locally without network access.

10.1 Exact arithmetic and independent enumeration

The public module `code/record_sum.py` implements the finite polynomial (2.1) with integer coefficients. It computes the diagonal coefficient at each stage. A separate calculation multiplies the

shifted product in exact rational arithmetic, then multiplies by $(n - 1)!$. The certificate compares these methods through $n = 60$ and checks the displayed initial values, including a_0 and the index shift.

A third calculation enumerates every permutation through $n = 8$. It separately computes the sum of record positions and the sum of cycle minima from the permutation itself. It does not use the product formulas to determine those statistics. This checks both interpretations, while the proofs in Section 2 establish the identities for all n .

For the rooted-tree cancellation, the exact coefficient of s^k in B_q is calculated as

$$[s^k]B_q(s) = \sum_{\substack{1 \leq r \leq k+1 \\ q|r}} \frac{r^{r-1}(-r)^{k-r+1}}{r!(k-r+1)!}. \quad (10.1)$$

This rational sum independently verifies cancellation through degree 40 at $q = 1$ and the filtered leading terms for orders 2 through 14. It verifies many finite coefficients of (4.5); the identity for every degree follows from the inverse-function argument, not from extrapolating these checks.

Symbolic calculations verify the logarithmic local jet at -1 , the exact transfers in (8.6), and the sign and constant changes under $m = n - 1$. They also check the simple-zero delay at $\pm i$. Tests of a general transfer identity are identified as such. The code does not generate fabricated higher sequence coefficients by feeding arbitrary amplitudes into a generic symbolic method.

10.2 High precision constants and directed coefficient bounds

The Gamma amplitudes at ω and i are evaluated at high precision and compared with independent polylogarithmic sums after isolating the $j = 1$ factor. A second computation at higher precision checks stability. The analytic truncation bound (8.10) is reported separately from special-function roundoff. The special-function evaluations use ordinary high precision arithmetic, not interval arithmetic, and the printed digits are not certified enclosures.

For optional larger finite-product calculations, the Decimal routine constructs lower and upper coefficient arrays. Since all coefficients, reciprocals $1/j$, and update operations are nonnegative, rounding each division, multiplication, and addition down gives a lower bound, while rounding them up gives an upper bound. Induction over the descending coefficient updates proves enclosure of the exact finite rational coefficients. The certificate compares small cases with exact fractions.

These directed intervals account for roundoff in the finite nonnegative product. They do not certify the error in an asymptotic expansion, the evaluation of γ or complex Gamma functions, or an inverse threshold. In particular, a small residual after subtracting (1.4) remains a diagnostic even if the coefficient itself has been rigorously enclosed. The optional large computation is quadratic in its cutoff and is not required for the default certificate.

10.3 Supported commands and bounded inputs

The package README gives the complete invocation syntax. Representative commands are

```
python -B code/checks.py
python -B -O code/checks.py
python -B code/record_sum.py exact -n 60
python -B code/record_sum.py roots -digits 80 -cutoff 115
```

```
python -B code/record_sum.py inverse -log-y 1000 -order 3
python -B code/record_sum.py decimal -n 1800 -digits 60
```

The inverse command accepts $\log y$ directly, avoiding construction of a huge target. It evaluates the actual $K = 2$ or $K = 3$ model in (9.3), with its analytic derivative and the real frequency convention. Fixed-count undamped Newton iterates and a residual-controlled numerical root are distinguished in its output. The latter is a numerical solution of the model, not a certified integer inverse of a_n .

The supported interface rejects out-of-range inputs, invalid precision requests, nonfinite numerical input, and explicit nonconvergence. Exact diagonal computation is bounded at $n = 200$. Optional Decimal computation allows $1 \leq n \leq 6000$ and $20 \leq \text{precision} \leq 200$. The numerical inverse permits $10 \leq \log y \leq 10^{12}$, with bounded precision and iteration counts. The documented decimal integer conversion cap is 640 digits. It is not disabled to accommodate unbounded inputs.

All program guards use explicit conditional failures, not Python assertions that disappear with optimization. Every child Python process is launched with `-B`; the environment also disables bytecode writes. The normal and optimized modes are checked against the same frozen JSON receipts. These finite guard tests verify the supported interface; they are not a general proof of security against every possible filesystem race or hostile interpreter.

10.4 A frozen reproducible archive

The public archive contains the editable TeX, bibliography and source ledger, Python code, dependency pins, arithmetic receipts, and the verified PDF. It contains no third-party paper PDFs and no private working audits. The manifest records a SHA-256 digest for every package file except the manifest itself. It detects changes, missing files, additions, unexpected directories, and symlinks; it is an integrity record, not a cryptographic signature from an external authority.

The build verifies the manifest before running the arithmetic and guard suites. It compiles a disposable source copy with a private TeX format and cache. Every TeX invocation, including private format generation and version inspection, uses `-no-shell-escape`. Fixed dates and a fixed trailer identifier make the reference PDF deterministic. The build rejects overfull boxes, missing glyphs, and unresolved references, and requires the regenerated PDF to match the frozen PDF byte for byte.

Output paths must be new and outside the source package, with an existing parent directory. Existing outputs, symlink ancestors, and raw dot or dot-dot path components are rejected. The build does not merge into an old directory or write generated files back into the source tree.

The ZIP has sorted entries, a fixed timestamp, fixed modes, and stored compression. A separate verifier first compares the actual ZIP with the trusted frozen package, then extracts it into two fresh directories and runs complete normal and optimized builds. It compares all archive members and metadata, the full ZIP, the PDF, the arithmetic and guard receipts, and the build receipt, and verifies that both extracted source trees remain unchanged. Build logs may contain temporary paths and are not claimed to be byte-identical.

Reference tool versions are pinned in the README and receipts. Exact PDF byte reproduction depends on the TeX engine, fonts, and package versions; a different toolchain is reported as a mismatch rather than silently accepted. The mathematical proof does not depend on a particular TeX release or on successful reproduction of a floating point decimal.

10.5 Explicit limits

For ease of interpretation, the report does not claim any of the following:

- convergence of an infinite root-of-unity transseries or uniformity as K grows;
- an effective numerical constant or starting index for the asymptotic errors;
- a finite threshold certificate obtained by rounding a numerical model root;
- interval certification of the printed complex Gamma amplitudes;
- a complete public implementation of every higher P_j in elementary constants;
- formal proof-assistant verification, peer review, or a worldwide priority search.

The assertions that are proved are the every-fixed-order expansion, the complete zero-frequency cancellation, the displayed low-order coefficients, the natural boundary and exceptional zeros, and the asymptotic integer envelopes with finite Newton centers.

[write] **The shipped files, 7 October 2026.** In ProveIt the PDF, the manifest and the delivered README are not shipped; the programs are `code/*.py` (the builder as `code/build.py`), and the receipts `code/receipt.json`, `code/guard_receipt.json` and the requirements are shipped under `data/`. The builder and the archive verifier look for the receipts under `code/` and for the manifest at the root, so they run only in a re-extracted archive; `checks.py` writes its receipt to standard output and runs from the shipped files. The report README gives commands.

11 Source and duplicate search boundaries

11.1 Sequence identification and freshness

The source reconnaissance dated 5 October 2026 obtained the complete indexed OEIS record for A368246, including its definition, offset, data, formula, examples, program, and cross-reference to A143946. It identifies the cycle-minimum diagonal, $a_0 = 1$, and the initial values used here. Direct fresh requests for A368246 and its internal entry were unsuccessful, including an HTTP 403 response. A newly verified live revision timestamp was therefore not obtained.

The related primary entry A143946 was retrieved and gives the record-position triangle and its row polynomial

$$z(z^2 + 1)(z^3 + 2) \cdots (z^n + n - 1).$$

The primary papers also ground the statistic independently of the indexed entry. The exact counting proof in Section 2 provides a further check that the normalization and index agree with this definition.

The indexed record includes a historical conjecture attributing the constant $e^{-\gamma}$ to Václav Kotěšovec on 29 December 2023. That is a statement about the inspected record, not evidence that the leading asymptotic remains open. As explained in Section 2, the 2013 local limit theorem already implies it. No current OEIS claim of an unresolved leading constant is made here.

11.2 The inspected mathematical literature

Louchard [4, Eq. (1)] explicitly supplies the finite polynomial, and Eq. (2) gives its probability normalization. His introductory local-limit discussion and Eq. (4) connect the statistic to the

Dickman distribution and credit earlier work. Kortchemski [3] treats permutation records and a different large-deviation regime. Neither the known finite polynomial nor the elementary infinite-product reduction is presented here as a novelty.

Giuliano, Szewczak and Weber [5, Theorem 2.1] provide the exact local limit statement used above. Their theorem, rather than a heuristic density argument, proves the leading constant on the diagonal. The stronger quantitative local estimates in de la Bretèche and Tenenbaum [6, Eqs. (2.14), (3.1)] were also inspected. Their displayed diagonal error is too large to identify the n^{-2} correction in b_n established here. A related oscillatory term in a total-variation analysis cannot simply be read as this sequence’s coefficient.

Flajolet, Fusy, Gourdon, Panario and Pouyanne [7, Theorems 1–2, Section 3] supply the classical hybrid framework. Their worked product is

$$\prod_{j \geq 1} \left(1 + \frac{z^j}{j} \right),$$

whereas ours has exponent $j + 1$. The unshifted example has a nonzero smooth correction sector; copying those coefficients would give the wrong answer for A368246. The use of Gamma products, local root expansions, and a natural boundary is part of that established method. The calculation specific to the shifted product is the identity (4.5) and its consequences.

11.3 Bounded duplicate checking

Current repository catalogues and exact-number searches were checked during reconnaissance. Actual near-match article sources on two Fourier peaks, path-forest expansions, partition orientation sums, a level-seven Gamma constant, and a cubic restricted-partition boundary were inspected rather than relying only on titles. Their models differ from the shifted record product. Exact-number searches for A368246 returned no matching article in the searched repository.

Searches in the available document collection also used the sequence number and semantic terms such as record positions, cycle minima, and Dickman asymptotics. Returned near matches were opened and identified as different models, including deterministic record walks and diagonal alignment counts. Search results were sometimes fuzzy. These checks support only the statement that no duplicate was found among the inspected material. They do not establish exhaustive absence from the collection, the internet, unpublished work, or the broader literature.

The editable `SOURCES.md` records source URLs and the scope of these checks. It deliberately omits private document identifiers and working-audit files. The result is a self-contained mathematical report with a bounded source ledger, not a certificate of priority.

Remark 11.1 ([write] The OEIS entries, 7 October 2026). The write read the entries in the internal format on 7 October 2026; the source could not (Section 11). *A368246*, revision #22 (4 January 2024), by Alois P. Heinz (18 December 2023): “Number of permutations of $[n]$ whose cycle minima sum to n .”, with the record-position comment, $a(n) = A143946(n, n)$, a Maple program, a b-file by Heinz for $n = 0, \dots, 451$, and the formula “ $a(n) \sim c * (n-1)!$, where $c = 0.561459\dots$, conjecture: $c = \exp(-\gamma) = A080130$, where γ is the Euler-Mascheroni constant A001620.”, signed by Václav Kotěšovec, 29 December 2023. *The conjecture, as posted, is proved*: Theorem 1.1 gives $b_n = A + O(n^{-2})$, hence $a_n \sim e^{-\gamma}(n-1)!$, and it already followed from Theorem 2.1 of Giuliano–Szewczak–Weber (2013), whose model (Z_k independent, equal to 1 with probability $1/k$, $T_n = \sum_{k \leq n} kZ_k$) is that of Section 2, as the source says; the printed 0.561459 is a correct

truncation of $e^{-\gamma} = 0.5614594835\dots$ *A143946*, revision #38 (13 March 2026), by Emeric Deutsch (21 September 2008), with the row polynomial $t(t^2 + 1)(t^3 + 2)\cdots(t^n + n - 1)$, as the source states. *Checks.* All 452 b-file terms of A368246 equal the write's evaluation of (2.1). No OEIS edit is part of this work.

12 Further research questions

The fixed-order theorem leaves several concrete questions. They are proposed directions rather than results proved by this report.

Effective constants and onset. The tail majorants in Section 3 are explicit, but the proof does not assemble numerical bounds for all local jets, their complementary arcs, and the final Fourier norm. Carrying out that assembly could produce certified constants and onsets in (1.2). Combined with interval evaluation of the finite real model, it could turn the asymptotic envelope into a practical finite threshold certificate. At present, the exact count comparison remains necessary whenever a numerical inverse is used without such bounds.

Certified higher amplitudes. The finite prescription in Section 7 determines every fixed coefficient. A reusable implementation could construct the actual higher P_j , rigorously bound derivatives of the smooth tail, collect the minimal periods, and expose cancellations among root amplitudes. It would be useful to determine which constants admit Gamma or other standard special-function expressions and which are most naturally represented by convergent series. The public code here implements exact tree jets and transfer checks, with the full sequence model displayed only through order three.

More general exponent patterns and weights. A natural extension is to products with periodic, nonconstant perturbations of the exponent $j + 1$, or with controlled periodic weights. The question is which patterns cause complete or partial cancellation of the positive-root logarithmic sector, and how they alter the root frequencies. The constant-shift case alone is not an unresolved characterization: for

$$F_d(z) = \prod_{j \geq 1} (1 + z^{j+d}/j), \quad d \in \mathbb{Z}_{\geq 0},$$

the positive logarithmic coefficient is $-T(se^{-ds})/s = -1 - (1 - d)s + O(s^2)$. Its pole coefficient is again A , so already the first smooth correction, in coefficient index m , is $A(1 - d)/m$. Thus full cancellation among constant nonnegative integer shifts can occur only at $d = 1$, the case proved here. More varied exponent patterns require additional analysis.

Large order and optimal truncation. Nothing in the fixed- K proof controls the growth of P_j or of its remainder constants as $j \rightarrow \infty$. Establishing that growth could address optimal truncation, interactions among increasing root orders, and the scale of the error beyond all retained algebraic terms. The natural boundary makes these questions especially sensitive to uniformity. No convergent infinite transseries, optimal truncation rule, or exponentially small remainder is asserted in this report.

[write] **Added 7 October 2026, under Vladimir's standing rule of 4 October 2026.** These four questions are the source's own and stay open; the constant-shift observation in the third is proved in the text

(and agrees numerically). [Independent check, 7 October 2026: “proved” is too strong. The text computes the positive-sector coefficient $-T(se^{-ds})/s$ for every d . The global transfer of Sections 3–7, which turns that sector into a coefficient statement, is written for $d = 1$ only, with its special treatment of the zeros of $1 + z^2$ at $\pm i$. For other d it is not repeated. There the only factor that can vanish on the unit circle is $1 + z^{1+d}$, a polynomial with simple zeros at the $(d + 1)$ -th roots of -1 , so the adaptation looks routine, but it has not been carried out. Read the observation as derived, not proved. Numerically, $m(b_m^{(d)} - A)/(A(1 - d))$ at $m = 1000, 2000$ and 3000 is $0.99228, 0.99578, 0.99705$ for $d = 0$, and $1.00417, 1.00244, 1.00176$ for $d = 2$. For $d = 3$ it is $1.0111, 1.0062, 1.0044$.] The write adds from the non-claims interval enclosures of the Gamma amplitudes and an implementation of the higher P_j . No claim of the source was found false; Kotěšovec’s conjecture in A368246 is proved (Remark 11.1).

References

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- [8] ProveIt, *Transseries and inversion*, Analysis/Transseries/docs/series-and-transseries/Transseries_And_Inversion/transseries_and_inversion.tex; Part 0 (p0:def:model, p0:thm:lambert-core, p0:def:three-inverses, p0:thm:staircase) and the chapter on the Lambert template (plt:thm:lw-template). [Added in the write.]